38.3: People modify crops by breeding and genetic engineering

People have intervened in the reproduction and genetic makeup of plants since the dawn of agriculture. Maize, for example, owes its existence to humans. Left on its own in nature, maize would soon become extinct for the simple reason that it cannot spread its seeds. Maize kernels are not only permanently attached to the central axis (the “cob”) but also permanently protected by tough, overlapping leaf sheathes (the “husk”) **(Figure 38.16)**. These attributes arose by artificial selection by humans. Despite having no understanding of the scientific principles underlying plant breeding, early farmers domesticated most of our crop species over a relatively short period about 10,000years ago.

Figure 38.16

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Maize: a product of artificial selection.
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Modern maize (bottom) was derived from teosinte (top). Teosinte kernels are tiny, and each row has a husk that must be removed to get at the kernel. The seeds are loose at maturity, allowing dispersal, which probably made harvesting difficult for early farmers. Neolithic farmers selected seeds from plants with larger cob and kernel size as well as the permanent attachment of seeds to the cob and the encasing of the entire cob by a tough husk.

HHMI BioInteractive Video

Popped Secret: The Mysterious Origin of Corn

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Plant Breeding

Plant breeding is the art and science of changing the traits of plants in order to produce desired characteristics. Breeders scrutinize their fields carefully and travel far and wide searching for domesticated varieties or wild relatives with desirable traits. Such traits occasionally arise spontaneously through mutation, but the natural rate of mutation is too slow and unreliable to produce all the mutations that breeders would like to study. Breeders sometimes hasten mutations by treating large batches of seeds or seedlings with radiation or chemicals.

In traditional plant breeding, when a desirable trait is identified in a wild species, the wild species is crossed with a domesticated variety. Generally, those progeny that have inherited the desirable trait from the wild parent have also inherited many traits that are not desirable for agriculture, such as small fruits or low yields. The progeny that express the desired trait are again crossed with members of the domesticated species and their progeny examined for the desired trait. This process is continued until the progeny with the desired wild trait resemble the original domesticated parent in their other agricultural attributes.

While most breeders cross-pollinate plants of a single species, some breeding methods rely on hybridization between two distant species of the same genus. Such crosses sometimes result in the abortion of the hybrid seed during development. Often in these cases the embryo begins to develop, but the endosperm does not. Hybrid embryos are sometimes rescued by surgically removing them from the ovule and culturing them *in vitro*.

It is important to note that the natural genetic modification of plants began long before humans started altering crops by artificial selection. For example, the wheat species we rely on for much of our food evolved by natural hybridization between different species of grasses. Such hybridization is common in plants and has long been exploited by breeders to introduce genetic variation for artificial selection and crop improvement. Another instance of natural genetic modification has been revealed by the genomic sequencing of sweet potato (*Ipomoea batatas*). Apparently an early ancestor of the modern sweet potato

came into contact with the soil bacterium *Agrobacterium*, and a horizontal gene transfer event (see Concept 26.6) occurred. Thus, the introduction of a **transgene**, a gene that has been transferred from one organism into another, can occur in nature as well as in a genetic engineering laboratory.

Plant Biotechnology and Genetic Engineering

Plant biotechnology has two meanings. In the general sense, it refers to innovations in the use of plants (or substances obtained from plants) to make products of use to humans—an endeavor that began in prehistory. In a more specific sense, biotechnology refers to the use of genetically modified (GM) organisms in agriculture and industry. Indeed, in the last few decades, genetic engineering has become such a powerful force that the terms *genetic engineering* and *biotechnology* have become synonymous in the media. The revolutionary CRISPR-Cas9 gene-editing technology (see Figure 20.14) is changing plant biology as fast as it is breaking ground in other fields.

Unlike traditional plant breeders, modern plant biotechnologists, using techniques of genetic engineering, are not limited to the transfer of genes between closely related species or genera. For example, traditional breeding techniques could not be used to insert a desired gene from daffodil into rice because the many intermediate species between rice and daffodil and their common ancestor are extinct. In theory, if breeders had the intermediate species, over the course of several centuries they could probably introduce a daffodil gene into rice by traditional hybridization and breeding methods. With genetic engineering, however, such gene transfers can be done more quickly, more specifically, and without the need for intermediate species.

In the remainder of this chapter, we explore the prospects and controversies surrounding the agricultural use of crops that are genetically modified organisms (GMOs). Advocates for plant biotechnology believe that the genetic engineering of crop plants is the key to overcoming some of the most pressing problems of the 21st century, including world hunger and fossil fuel dependency.

Interview

Interview with Luis Herrera-Estrella: Using GMOs to help farmers

Reducing World Hunger and Malnutrition

Although global hunger affects nearly a billion people, there is much disagreement about its causes. Some argue that food shortages arise from inequities in distribution and that the most poverty-stricken simply cannot afford food. Others regard food shortages as evidence that the world is overpopulated—that the human species has exceeded the carrying capacity of the planet (see Concept 53.3). Whatever the causes of malnutrition, increasing food production is a humane objective. Because land and water are the most limiting resources, the best option is to increase yields on already existing farmland. Indeed, there is very little "extra" land that can be farmed, especially if the few remaining pockets of wilderness are to be preserved. Plant biotechnology can help make these crop yields possible.

Crops that have been genetically modified to express transgenes from *Bacillus thuringiensis,* a soil bacterium, require less pesticide. The transgenes involved encode a protein (*Bt* toxin) that is toxic to many insect pests **(Figure 38.17)**. The *Bt* toxin used in crops is produced in the plant as a harmless protoxin that only becomes toxic if activated by alkaline conditions, such as in the guts of most insects. Because vertebrates have highly acidic stomachs, protoxin consumed by humans or livestock is rendered harmless by denaturation.

Figure 38.17

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Non-*Bt* versus *Bt* maize.
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Field trials reveal that non-Bt maize (left) is heavily damaged by insect feeding and Fusarium mold infection, whereas Bt maize (right) suffers little or no damage.

Biofortification, improving the nutritional quality of plants, is another strategy in the war on world hunger. For example, tens of thousands of children lose their sight each year because of vitamin A deficiencies. More than half of these children die within a year of becoming blind. In response to this crisis, genetic engineers created "Golden Rice," a transgenic variety supplemented with transgenes that enable it to produce grain

with increased levels of beta-carotene, a precursor of vitamin A. In 2021, after decades of safety testing and legal challenges, Golden Rice was finally approved for cultivation by the Philippines: massive production, focusing on provinces with high vitamin A deficiency, commenced in 2022.

Another major target for biofortification improvement is cassava (*Manihot esculenta*), a starchy root crop that provides carbohydrates for 800 million of the world's most impoverished people **(Figure 38.18)**. Researchers are also engineering plants with enhanced resistance to disease. In one case, a transgenic papaya that is resistant to a ringspot virus was introduced into Hawaii, thereby saving its papaya industry.

Figure 38.18

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Engineering the perfect food?
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Cassava grows well in poor dry soils but must be supplemented by consumption of other crops. If those crops fail, people suffer malnutrition. What if cassava alone could supply a balanced diet? Transgenic, biofortified cassava plants have been developed with greatly increased levels of iron and beta-carotene (a vitamin A precursor). Other improvements have been made in the size of its storage roots, its ease of processing (less cyanide-producing chemicals), and its resistance to cassava mosaic virus disease.

Interview

Interview with Dennis Gonsalves: Saving the papaya industry

Considerable controversy has arisen concerning transgenic crops that are resistant to the herbicide glyphosate. Glyphosate is lethal to a wide variety of plants because it inhibits a key enzyme in a biochemical pathway that is found in plants (and most bacteria) but not in animals. Researchers discovered a bacterial strain that had undergone a mutation in the gene encoding this enzyme that rendered it glyphosate resistant. When this mutated bacterial gene was spliced into the genome of various crops, these crops also became glyphosate resistant. Farmers achieved almost total weed control by spraying glyphosate over their fields of glyphosate-resistant crops. Unfortunately, the overuse of glyphosate created a huge selective pressure on weed species, with the result that resistance to glyphosate has evolved in many of them. There has also been a growing appreciation in recent decades of the role that gut bacteria play in animal and human health, and claims have been made that glyphosate may be having negative effects on the health of humans and livestock by interfering with beneficial gut bacteria. Glyphosate has also been the subject of many studies and legal challenges regarding its potential to cause cancer. Although billions of dollars in damages have been awarded in court cases alleging links between glyphosate-based weed killers and cancer, the research is conflicting.

Reducing Fossil Fuel Dependency

Global sources of inexpensive fossil fuels, particularly oil, are rapidly being depleted. Moreover, most climatologists attribute global warming mainly to the rampant burning of fossil fuels, such as coal and oil, and the resulting release of the greenhouse gas CO_2. How can the world meet its energy demands in the 21st

century in an economical and nonpolluting way? In certain localities, wind or solar power may become economically viable, but such alternative energy sources are unlikely to fill the global energy demands completely. Many scientists predict that **biofuels**—fuels derived from living biomass—could produce a sizable fraction of the world's energy needs in the not-too-distant future. **Biomass** is the total mass of organic matter in a group of organisms in a particular habitat. The use of biofuels from plant biomass would reduce the net emission of CO_2. Whereas burning fossil fuels increases atmospheric CO_2 concentrations, biofuel crops reabsorb by photosynthesis the CO_2 emitted when biofuels are burned, creating a cycle that is carbon neutral.

In working to create biofuel crops from wild precursors, scientists are focusing their domestication efforts on fast-growing plants, such as switchgrass (*Panicum virgatum*) and poplar (*Populus trichocarpa*), that can grow on soil that is too poor for food production. Scientists do not expect the plant biomass to be burned directly. Instead, the polymers in cell walls, such as cellulose and hemicellulose, which constitute the most abundant organic compounds on Earth, would be broken down into sugars by enzymatic reactions. These sugars, in turn, would be fermented into alcohol and distilled to yield biofuels. In addition to increasing plant polysaccharide content and overall biomass, researchers are trying to genetically engineer the cell walls of plants to increase the efficiency of the enzymatic conversion process.

The Debate over Plant Biotechnology

Much of the debate about GMOs in agriculture is political, social, economic, or ethical and therefore outside the scope of this book. But we *should* consider the biological concerns about GM crops. Some biologists, particularly ecologists, are concerned about the unknown risks associated with the release of GMOs into the environment. The debate centers on the extent to which GMOs could harm the environment or human health. Those who want to proceed more slowly with agricultural biotechnology (or end it) are concerned about the unstoppable nature of the "experiment." If a drug trial produces unanticipated harmful results, the trial is stopped. But we may not be able to stop the "trial" of introducing novel organisms into the biosphere. Here we examine some criticisms that have been leveled by opponents of GMOs, including the alleged effects on human health and nontarget organisms and the potential for transgene escape.

Issues of Human Health

Many GMO opponents worry that genetic engineering may inadvertently transfer allergens, molecules to which some people are allergic, from a species that produces an allergen to a plant used for food. However, biotechnologists are already removing genes that encode allergenic proteins from soybeans and other crops. So far, there is no credible evidence that GM plants designed for human consumption have allergenic effects on human health. In fact, some GM foods are potentially healthier than non-GM foods. For example, *Bt*

maize (the transgenic variety with the *Bt* toxin) contains 90% less of a fungal toxin that causes cancer and birth defects than non-*Bt* maize. Called fumonisin, this toxin is highly resistant to degradation and has been found in alarmingly high concentrations in some batches of processed maize products, ranging from cornflakes to beer. Fumonisin is produced by a fungus (*Fusarium*) that infects insect-damaged maize. Because *Bt* maize generally suffers less insect damage than non-GM maize, it contains much less fumonisin.

Assessing the impact of GMOs on human health also involves considering the health of farmworkers, many of whom were commonly exposed to high levels of chemical insecticides prior to the adoption of *Bt* crops. In India, for example, the widespread adoption of *Bt* cotton has led to a 41% decrease in insecticide use and an 80% reduction in the number of acute poisoning cases involving farmers.

Possible Effects on Nontarget Organisms

Many ecologists are concerned that GM crops may have unforeseen effects on nontarget organisms. One laboratory study indicated that the larvae (caterpillars) of monarch butterflies responded adversely and even died after eating milkweed leaves (their preferred food) heavily dusted with pollen from transgenic *Bt* maize. This study has since been discredited, affording a good example of the self-correcting nature of science. As it turns out, when the original researcher shook the male maize inflorescences onto the milkweed leaves in the laboratory, the filaments of stamens, opened microsporangia, and other floral parts also rained onto the leaves. Subsequent research found that it was these other floral parts, *not* the pollen, that contained *Bt* toxin in high concentrations. Unlike pollen, these floral parts would not be carried by the wind to neighboring milkweed plants when shed under natural field conditions. Only one *Bt* maize line, accounting for less than 2% of commercial *Bt* maize production (and now discontinued), produced pollen with high *Bt* toxin concentrations.

In considering the negative effects of *Bt* pollen on monarch butterflies, one must also weigh the effects of an alternative to the cultivation of *Bt* maize—the spraying of non-*Bt* maize with chemical pesticides. Subsequent studies have shown that such spraying is much more harmful to nearby monarch populations than is *Bt* maize production. Although the effects of *Bt* maize pollen on monarch butterfly larvae appear to be minor, the controversy has emphasized the need for accurate field testing of all GM crops and the importance of targeting gene expression to specific tissues to improve safety.

Addressing the Problem of Transgene Escape

Perhaps the most serious concern raised about GM crops is the possibility of the introduced genes escaping from a transgenic crop into related weeds through crop-to-weed hybridization. The fear is that the spontaneous hybridization between a crop engineered for herbicide resistance and a wild relative might give rise to a “superweed” that would have a selective advantage over other weeds in the wild and would be much more difficult to control in the field. GMO advocates point out that the likelihood of transgene escape

depends on the ability of the crop and weed to hybridize and on how the transgenes affect the overall fitness of the hybrids. A desirable crop trait—a dwarf phenotype, for example—might be disadvantageous to a weed growing in the wild. In other instances, there are no weedy relatives nearby with which to hybridize; soybean, for example, has no wild relatives in the United States. However, canola, sorghum, and many other crops do hybridize readily with weeds, and crop-to-weed transgene escape in a turfgrass has occurred. In 2003 a transgenic variety of creeping bentgrass (*Agrostis stolonifera*) genetically engineered to resist the herbicide glyphosate escaped from an experimental plot in Oregon following a windstorm. Despite efforts to eradicate the escapee, 62% of the *Agrostis* plants found in the vicinity three years later were glyphosate resistant. So far, the ecological impact of this event appears to be minor, but that may not be the case with future transgenic escapes.

Many strategies are being pursued with the goal of preventing transgene escape. For example, if male sterility could be engineered into plants, these plants would still produce seeds and fruit if pollinated by nearby nontransgenic plants, but they would produce no viable pollen. A second approach involves genetically engineering apomixis into transgenic crops. When a seed is produced by apomixis, the embryo and endosperm develop without fertilization. The transfer of this trait to transgenic crops would therefore minimize the possibility of transgene escape via pollen because plants could be male-sterile without compromising seed or fruit production. A third approach is to engineer the transgene into the chloroplast DNA of the crop. Chloroplast DNA in many plant species is inherited strictly from the egg, so transgenes in the chloroplast cannot be transferred by pollen. A fourth approach for preventing transgene escape is to genetically engineer flowers that develop normally but fail to open. Consequently, self-pollination would occur, but pollen would be unlikely to escape from the flower. This solution would require modifications to flower design. Several floral genes have been identified that could be manipulated to this end.

The continuing debate about GMOs in agriculture exemplifies one of this textbook's recurring ideas: the relationship of science and technology to society. Technological advances almost always involve some risk of unintended outcomes. In the case of genetically engineered crops, zero risk is probably unattainable. Therefore, scientists and the public must assess on a case-by-case basis the possible benefits of transgenic products versus the risks that society is willing to take. The best scenario is for these discussions and decisions to be based on sound scientific information and rigorous testing rather than on reflexive fear or blind optimism.